MATHS-QE-01

Topics Covered: *Polynomials Relations between roots and coefficient Formation of quadratic equations with given roots*

Total Count of Questions: *22*

Expected Time of Attempt by Students: *120 min*

1. The product of the real roots of the equation $|2x + 3|^2 - 3|2x + 3| + 2 = 0$, is
(a) 5/4 (b) 5/2
(c) 5 (d) 2

2. The roots of the equation $|x^2 - x - 6| = x + 2$ are
(a) –2, 1, 4 (b) 0, 2, 4
(c) 0, 1, 4 (d) –2, 2, 4

3. If (1 – p) is a root of quadratic equation $x^2 + px + (1 - p) = 0$, then its roots are
(a) 0, –1 (b) –1, 1
(c) 0, 1 (d) –1, 2

4. Product of real roots of the equation
$x^2 + |x| + 9 = 0$
(a) is always positive (b) is always negative
(c) does not exist (d) none of the above

5. The integral value of x satisfing $|x^2 + 4x + 3| + 2x + 5 = 0$ is
(a) –4 (b) –3
(c) –2 (d) –1

6. If p, q are the roots of the equation $x^2 + px + q = 0$ where both p and q are non-zero, then (p, q) =
(a) (1, 2) (b) (1, –2)
(c) (–1, 2) (d) (–1, –2)

7. If the equation $(k - 2)x^2 - (k - 4)x - 2 = 0$ has difference of roots as 3 then the value of k is
(a) 1, 3 (b) 3, 3/2
(c) 2, 3/2 (d) 3/2, 1

8. The roots of the equation $x^2 + px + q = 0$ are $\tan 22^\circ$ and $\tan 23^\circ$ then
(a) p + q = 1 (b) p + q = –1
(c) p – q = 1 (d) p – q = –1

9. If α, β are the roots of the equation $x^2 - p(x + 1) - c = 0$, then $(\alpha + 1)(\beta + 1) =$
(a) c (b) c–1
(c) 1–c (d) none of these

10. If the difference between the roots of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is same and $a \neq b$, then
(a) a + b + 4 = 0 (b) a + b –4 = 0
(c) a – b –4 = 0 (d) a – b + 4 = 0

11. If roots of the equation $x^2 + ax + 25 = 0$ are in the ratio of 2 : 3 then the value of a is
(a) $\frac{\pm 5}{\sqrt{6}}$ (b) $\frac{\pm 25}{\sqrt{6}}$
(c) $\frac{\pm 5}{6}$ (d) none of these

12. The sum of all real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$, is

13. The sum of the roots of the equation, $x^2 + |2x - 3| - 4 = 0$, is $\sqrt{2}k$. Then value of *k* is

14. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has how many real roots

15. If α and β are the roots of $4x^2 + 3x + 7 = 0$, then the value of $\frac{1}{\alpha^3} + \frac{1}{\beta^3}$ is of the form $\frac{225}{P}$. Then find the value of P.

16. If α and β are the roots of $x^2 - P(x + 1) - C = 0$, then the value of $\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + C} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + C}$ is

17. If the roots of the equations $x^2 + 3x + 2 = 0$ & $x^2 - x + \lambda = 0$ are in the same ratio of $\frac{4\lambda}{9}$ then the value of λ is given by

18. The value of a for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other, is of the form $\frac{2}{3k}$. Find the value of *k*.

19. (i) Solve the equation: $\frac{x^2 - 3x}{x^2 - 1} + 2 + \frac{1}{x - 1} = 0$

(ii) Solve the equation: $1 - \frac{3 - 2x}{5 - x} = \frac{3}{3 - x} - \frac{x + 3}{x + 1}$

20. Find a, if $ax^2 - 4x + 9 = 0$ has integral roots.

21. (i) Find the values of a for which one of the roots of $x^2 + (2a + 1)x + (a^2 + 2) = 0$ is twice the other root. Find also the roots of this equation for these values of a.
(ii) If the difference of roots of the equation $2x^2 - (a + 1)x + a - 1 = 0$ is equal to their product, then prove that a = 2.

22. (i) If α, β are roots of the equation $x^2 - px + q = 0$, find the value of
(a) $\alpha^2(\alpha^2\beta^{-1} - \beta) + \beta^2(\beta^2\alpha^{-1} - \alpha)$
(b) $(\alpha - p)^{-4} + (\beta - p)^{-4}$
(ii) If α, β are roots of $ax^2 + bx + c = 0$, then find the value of $(a\alpha + b)^{-3} + (a\beta + b)^{-3}$.

ANSWERS

1. (b) **2.** (d) **3.** (a) **4.** (c) **5.** (a)

6. (b) **7.** (b) **8.** (d) **9.** (c) **10.** (a)

11. (b) **12.** (4) **13.** (1.00) **14.** (0) **15.** (343)

16. (1) **17.** (2) **18.** (1) **19.** $(i)\ x = \frac{-1}{3}$ $(ii)\ -9, 2$ **20.** $\frac{-1}{4}, \frac{1}{3}$

21. $(i)\ a = 4; -1, -3$

22. $(i)(a)\ \frac{p\left(p^2-4q\right)\left(p^2-q\right)}{q}$

$(b)\ \frac{p^4-4p^2q+2q^2}{q^4}$ $(ii)\ \frac{b^3-3abc}{a^3b^3}$